

Build-Bench Challenge

Autonomous LLM Agents for Cross-Architecture Package Repair

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A working package can fail after migration

Cross-architecture repair connects source code, dependencies and packaging.



Maintainer effort

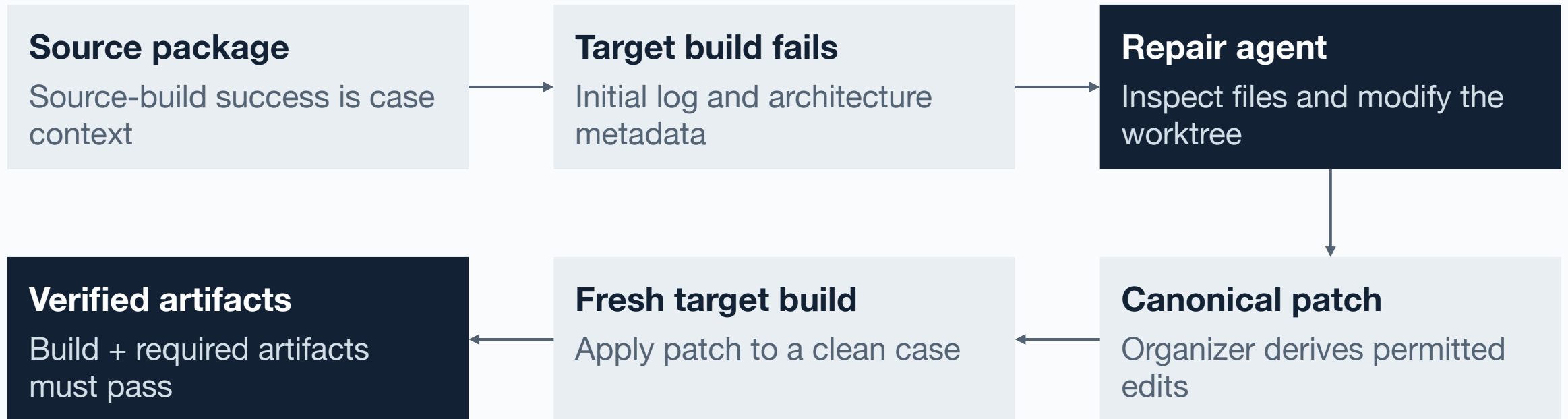
Trace the failure across unfamiliar logs, source and build files.

Our objective

Automate diagnosis and repair.
Confirm the result in a clean build.

A clean target build determines success

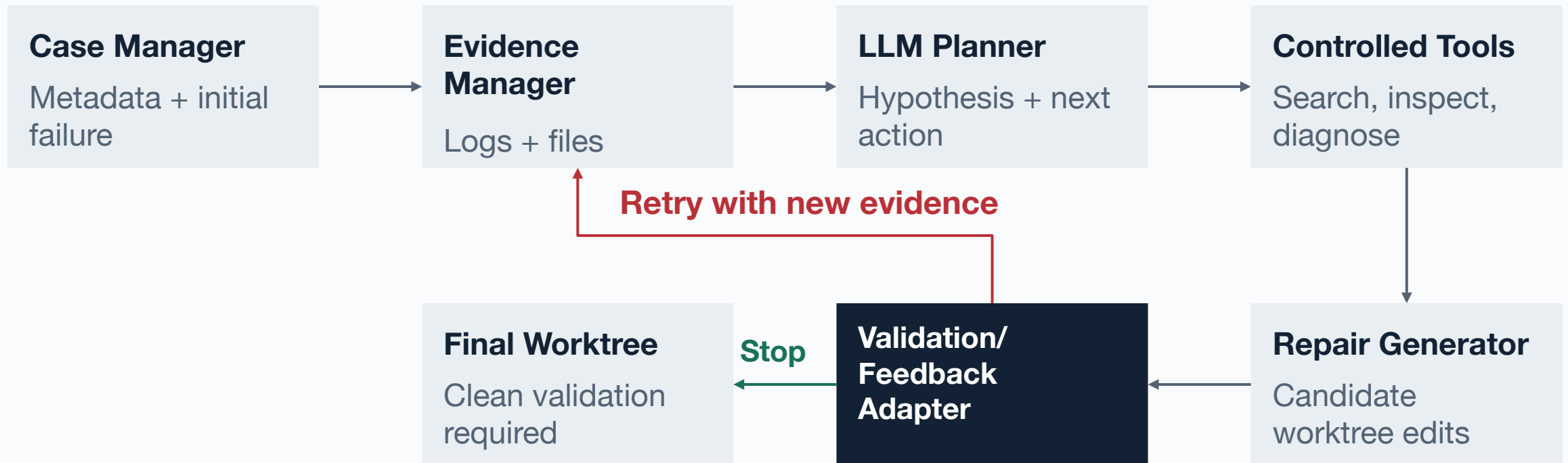
The organizer derives the patch and checks the resulting package artifacts.



Public: x86_64 and ARM64, both directions. Competition: also RISC-V

Proposed repair agent architecture

In-loop feedback depends on the interface. Final clean-build validation is required.



History/Metrics records outcomes. Budget Controls enforce resource limits.

Each attempt tests a specific hypothesis

- 01** **Observe**
- |
- 02** **Diagnose**
- |
- 03** **Gather evidence**
- |
- 04** **Repair**
- |
- 05** **Validate**

Bounded context

One libyuv log is 2.37 GB.
Select diagnostic windows and files.

Memory between attempts

Keep hypotheses, diffs and failures.
Stop repeated unsuccessful edits.

Initial research caps

5 candidates / 30 min / 30k tokens

Proposed caps, subject to stricter organizer limits. Success or repetition can stop earlier.

We will compare verified repairs and cost

PRIMARY METRIC

Verified Build Success Rate

Verified repairs $\div$ all cases in the frozen evaluation set

Fair comparison

Same cases, model and budget ceilings
Official baseline where reproducible
Failures remain in the denominator

Cost and failure records

Wall time and model tokens
Repair attempts and tool calls
Direction and failure breakdowns

Semester milestones have checkable outcomes

October 21

Demo 2

Runnable LLM prototype
10 fixed public cases
Baseline + per-case records

November 16

Demo 3

Same 10 cases, 3 runs
+1 mean repair, OR
20% fewer model tokens
preserving repaired cases

December 9

Final

40-case evaluation
30 held out from tuning
Reproducible agent,
commands and results

Competition freeze: November 13, before Demo 3. Qualification is conditional.

Evaluation targets. Repeat all 10 cases, hold budgets fixed and report every regression.

Current status and next steps

CURRENT STATUS

Dataset inspected: 200 cases across 188 packages.

Organizer's hello example reproduced on Linux.

Our LLM repair agent and public-case evaluation are scheduled for Demo 2.

Reproducible cases

Provision AWS and pin configs.
Reproduce a Debian failure.

Agent integration

Confirm model and feedback APIs.
Implement the bounded repair loop.

Core challenge: choose useful evidence and revise diagnoses within budget.